

# VIGNESH R

Python Developer | AI/ML Engineer | AI Engineer

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## PROFESSIONAL SUMMARY

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**Python Developer** with hands-on experience building scalable web applications, RESTful APIs, and AI-powered solutions using **Python, Django, Flask, FastAPI, JavaScript, TypeScript, React.js, Next.js, PostgreSQL, MySQL, and Supabase**. Proficient in backend architecture, database design, authentication, role-based access control, and API integration. Experienced in designing **Retrieval-Augmented Generation (RAG) systems**, integrating **Large Language Models (Nvidia API/Ollama)**, implementing **semantic search with ChromaDB and vector embeddings**, and developing modular, production-ready AI applications with clean architecture, secure data validation, and document processing pipelines.

Experienced professional transitioning from Payroll Operations and Financial Analysis to Software Engineering. Bringing 2.5+ years of expertise in financial data processing and process automation with transferable skills in data-driven problem-solving, systematic workflow optimization, and cross-functional collaboration. Proven ability to automate payroll tracking systems that cut processing time by 35% and reduced payroll errors by 25%, and eager to contribute to the software engineering industry by building scalable, data-driven applications and user-centered solutions.

## TECHNICAL SKILLS

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**Languages & Frameworks:** Python, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3

**Python Frameworks & Libraries:** Django, Flask, FastAPI, Django REST Framework (working knowledge)

**AI & LLM:** Retrieval-Augmented Generation (RAG), Large Language Models (Kimi API, Ollama), Prompt Engineering, Semantic Search, Vector Embeddings, ChromaDB, Sentence Transformers, PDF Document Processing, Text Chunking

**Frontend:** React.js, Next.js, Tailwind CSS, Zustand, responsive design

**Backend & APIs:** RESTful API Design & Development, Third-Party API Integration, Webhook-based Workflows, OpenAI-Compatible APIs, OData-style Data Fetching

**Databases:** PostgreSQL, MySQL, Supabase, ChromaDB, MongoDB (Learning), Database Schema Design, Query Optimization

**Architecture & Security:** Clean Architecture principles, role-based access control (RBAC), secure data validation, field-level data handling, RLS (Row-Level Security)

**DevOps & Tooling:** Git, GitHub (version control, repository management, code merges), VS Code, Agile delivery methodology

**AI-Assisted Development:** Claude Code, Cursor, Antigravity, Codex

**Automation & Workflow:** Process automation, dashboard automation, structured reporting pipelines, business process optimization

## FREELANCE PROJECTS

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### Physio Glides Physiotherapy ([Link](#))

*Tech Stack:* HTML5 | CSS3 | JavaScript | Bootstrap | Hosting | Antigravity | Figma

Developed a responsive physiotherapy clinic website with appointment booking functionality.

Integrated WhatsApp booking workflow, Google Maps, and service catalog features.

### NH Ads – Advertising Agency Website

*Tech Stack:* React.js | Next.js | Tailwind CSS | Figma | Claude | Codex | Antigravity

Designed and developed a responsive agency website from wireframes to deployment.

Built SEO-friendly frontend architecture with optimized performance and responsive layouts.

### HR Management Web App – Self Project ([GitHub Link](#))

*Tech Stack:* React.js | Node.js | Express.js | Supabase | PostgreSQL | Zustand | Figma | Claude | Codex | Antigravity

Developing a role-based HRMS platform for employee management, attendance tracking, and workflow automation.

Built scalable Zustand store with structured backend architecture and secure data validation.

### AI Personal Portfolio Chatbot

*Stack:* Python | FastAPI | Uvicorn | Ollama | Retrieval-Augmented Generation (RAG) | ChromaDB | Git | GitHub

Developed an AI-powered portfolio chatbot using **Python, FastAPI, ChromaDB, and Retrieval-Augmented Generation (RAG)** to answer user queries from resume and portfolio documents with semantic search.

Building a role-based dashboard architecture serving three distinct user types with differentiated data access rules directly applying backend authorization and access-control patterns relevant to **Python API** development.

**Ascendra – Team Management Platform (In Development)**

Stack: React Native | Supabase | AI Integrated

Designing a leader-first data ownership model with leader\_id-based Row-Level Security across all database tables.

Building a role-based dashboard architecture serving three distinct user types with differentiated data access rules directly applying backend authorization and access-control patterns relevant to Python API development.

**Achievement**

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Independently designed and shipped three live freelance/personal web projects (Physio Glides, NH Ads, HRMS platform) end-to-end — from architecture and UI to deployment — while working full-time in finance operations.

**WORK EXPERIENCE**

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**CogNethro • Process Associate • Jan 2026 – Feb 2026** *Chennai, India*

Supported HR screening and background verification for 200+ employees, maintaining 100% payroll-ready and audit-compliant records.

Managed benefits, reimbursements, and compliance data, improving payroll accuracy by 20% through systematic data validation workflows.

Leveraged strong communication skills to collaborate with cross-functional HR and Finance teams, optimizing payroll workflows and saving 20+ man-hours per month.

Built Excel payroll dashboards with dynamic expense and TDS tracking, enabling real-time financial reporting for senior management.

**RR Donnelley • Financial Associate • Aug 2023 – Oct 2025** *Chennai, India*

Executed end-to-end payroll processing — including TDS, deductions, garnishments, and 401(k) contributions — with full statutory tax compliance for 500+ employees.

Reduced payroll errors by 25% through rigorous reconciliation, validation controls, and audit-ready documentation using advanced Excel techniques.

Automated payroll tracking systems using Excel (VBA macros and pivot dashboards), cutting processing time by 35% and boosting team productivity across sprint cycles.

**CERTIFICATIONS**

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Certified UI/UX Designer – GUVI ZEN Class Full-Time Program (Feb 2024 – Oct 2024) | Figma, User Research, Wireframing, Prototyping, Usability Testing, Design Thinking

Full Stack Developer – Actively Pursuing at SLA | React.js, Python, Django, MySQL, APIs

Microsoft PowerPoint & Python – GUVI Self-Paced Certification

**EDUCATION**

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**Bachelor of Computer Applications (BCA)**

Vels Institute of Science, Technology and Advanced Studies | CGPA: 7.2 | June 2021 – May 2023

**Diploma in Computer Technology**

Meenakshi Krishnan Polytechnic College | 82% | April 2018 – May 2021

**ADDITIONAL INFORMATION**

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Languages Known: English (Professional Proficiency) | Tamil (Native)

Availability: Immediate Joiner | Open to Full-time, Hybrid & Remote Opportunities